

# GENERAL THETA FOUNDATIONS I: CAUSAL EXPERIMENTS, PREDICTIVE QUOTIENTS, AND RESOURCE-AWARE REDUCTION

QIAN QI

**ABSTRACT.** We study finite-resource realizations of predictive quotients of causal experiments. A structural theorem on admissible predictive trees identifies when a static adaptive quantization profile is achievable by a finite causal register at the same order. Strict decrease of predictive energy under prefix attachment makes every greedy tree suffix closed. The same finite tree supports both deletion after an acquisition and insertion of successive observations; all its vertices, including unary vertices, are charged. Consequently the theorem applies both to Markov-renewal orbit prediction and to a continually observed hidden shift with an infinite-coordinate posterior. We also construct finite recursive approximations of contracting measure-valued filters, with numerical and representation errors explicitly separated. Finally, a positive second-order reconstruction and a decision-theoretic cardinality converse give the sharp  $M^{-2/r}$  finite-experiment approximation order for a nondegenerate Gaussian shift. For a physical Poisson count-to-contact experiment this order holds jointly for  $M \lesssim N^{r/6}$ . The foundational statements specify the positive experiment, the future-test quotient, the causal simulation, and the resource transformation used in each application.

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## 1. INTRODUCTION

A statistic that preserves present predictions need not admit a finite recursive implementation. Conversely, a small implementable register need not preserve the decision problems of the underlying experiment. The purpose of this paper is to relate these three objects: a predictive quotient, its attainable geometry, and a charged causal realization. The order of the argument is

$$\begin{aligned} &\text{positive causal experiment} \longrightarrow \text{future-test quotient} \\ &\longrightarrow \text{causal simulation} \longrightarrow \text{finite-resource realization.} \end{aligned}$$

The quotient is always relative to a declared family of executable future tests. Resource preservation is always relative to a declared resource signature. Neither a measurable re-encoding by real numbers nor a terminal randomization is, by itself, a finite-memory implementation.

The principal structural result is Theorem 4.1. It starts with a factorial language, a stationary law, and a positive energy on its finite words. The hypotheses are local child comparisons, strict energy decrease under prefix attachment, bounded unary chains, and geometric separation of the predictive images. No quantization dimension is an input. If  $G_M$  is the energy of the greedy cut with at most  $M$  leaves, then its static squared quantization error is comparable to  $G_M$ . Every greedy tree is suffix closed. With  $L$  leaves it has at most  $(U + 1)(2L - 1)$  vertices, where  $U$  bounds unary chains. These vertices support two different causal operations. Deleting the oldest available future symbol realizes a renewal orbit code; prepending the newest observed symbol realizes a code of the entire observed past. Both operations have the same-order risk as the unrestricted static problem. The lower bounds allow randomized, time-dependent observers.

The two orientations are not merely two descriptions of one reset model. The Markov-renewal verification in Section 5 includes forbidden transitions and noninjective, delay-identifying observations. The second verification is a noisy hidden shift register whose tail persists at every time. Its complete posterior has infinitely many observation-dependent coordinates. Theorem 5.3 gives a matched whole-register excess-risk law in that experiment, including its noise dependence. Thus one structural realization theorem has a regenerative and a genuinely continually observed application.

A complementary approximation theorem treats predictive quotients that are measure valued rather than tree separated. Theorem 6.2 constructs a finite recursive register for contracting filters on  $\mathcal{P}([0, 1])$ . Its codebook has at most  $4^n$  states and error at most  $3/(2n(1 - \rho))$ , apart from initialization and explicitly charged numerical defects. The nonlinear example has no fixed-step common minorizer and no closed recursion of any prescribed finite list of polynomial moments. This is an upper approximation theorem, not a claim that every nonlinear filter has this contraction or that its logarithmic budget rate is universally optimal.

The experiment-comparison application is two-sided. Theorem 7.1 proves that an  $r$ -dimensional nondegenerate Gaussian location experiment, over a compact parameter set containing a ball, has optimal  $M$ -outcome approximation error  $\Theta(M^{-2/r})$  in Le Cam distance. The upper bound uses deterministic cells and positive second-order reconstruction, not first-order constant histograms. The lower bound applies to every experiment on at most  $M$  outcomes. For the actual local count-to-contact protocol of A2, Theorem 8.1 proves

$$\Delta(\mathcal{E}_{N,J}, \mathcal{G}) \leq C\{N^{-1/2} + JN^{-1/2} + J^{-2}\}, \quad |\mathcal{E}_{N,J}| \leq J^r.$$

It follows that  $\Delta \asymp M^{-2/r}$  throughout the growing-budget window  $1 \ll M \lesssim N^{r/6}$ . The proof includes the local normal, variance replacement, boundary crossing, reconstruction, and decision comparison arguments. The physical statistic is deterministic throughout.

**1.1. Relation to previous work.** The existence of path laws, conditional-probability quotients, Bayes orthogonality, information loss under coarsening, and the composition of statistical randomizations are classical; see [2, 1, 11, 12]. Sections 2–3 give the forms needed here, with the simulator and its persistent state included. We do not claim novelty for these identities or for their categorical notation.

Static graph-directed quantization is a particularly close antecedent. Kesseböhmer–Zhu [9], Theorem 1.1, determine Markov-type quantization dimension by the spectral radius of a matrix of  $(p_{ij}c_{ij}^r)^{s/(s+r)}$  and characterize finiteness of the upper coefficient through maximal strongly connected components. Their spatial theorem is not a theorem about a causal register. Our tree argument separates that familiar static mechanism from the additional suffix-closure condition and proves realization in both time orientations. The Markov application retains the renewal-weighted orbit and survival contribution; its spatial pressure is not a new discovery of graph-directed quantization. The earlier comparisons with functional quantization, dynamical rate distortion and zero-delay coding remain relevant [5, 3, 4]. A finite channel alphabet does not generally bound the entire persistent state of its coding policy.

Demirci–Kara–Yüksel [10], Theorems 3.5–3.6 of the cited preprint, obtain quantized-belief and finite-window policy approximations in their contraction-based control analysis. Our measure-valued result uses the classical contraction-plus-quantization mechanism, but provides a concrete entire-register cardinality and a same-observation recursive error bound for a specified nonlinear experiment. Its objective is prediction and terminal experiment simulation, not average-cost control optimality. The sharp hidden-shift theorem is instead a two-sided predictive-tree result. Neither theorem purports to subsume general partially observed control.

The new contribution is therefore not “finite memory”, “pressure”, or “Bayes reduction” in isolation. It is the combination of a checkable energy criterion, exact two-orientation realization, unrestricted register converses, and resource-typed experiment comparison. The Gaussian/A2 application supplies a separate decision-theoretic cardinality obstruction and an attainable joint sample/budget law. No priority claim for every component of these arguments is required.

**1.2. Reading convention.** This article is the canonical seventh revision of Volume I. Its proofs are self-contained at the level of the stated models. A separate *Complete development* contains this article followed by the unchanged earlier theorem bodies, including pressure, critical renewal, calibration, collision, and control results. The companion is a preservation and extended-reference object, not an alternative definition of the main paper. Source identities, historical consultation, and responses to the referee are recorded outside the mathematical text.

## 2. POSITIVE EXPERIMENTS AND PREDICTIVE QUOTIENTS

**2.1. Actual histories.** All spaces of states, observations and actions are standard Borel. For an unknown parameter  $\lambda$ , a controlled experiment has a probability kernel

$$K_t^\lambda(dx', dy, dc \mid x, a), \quad (2.1)$$

where  $c$  is the cost of the actual step. Failure, rejection and stopping are outcomes, not missing probability mass. A policy chooses  $a_t$  from the observed past and independent randomization. Preparation has a specified law and cost. With an absorbing symbol after stopping, repeated kernel integration constructs one probability law on the infinite history space. Its finite marginals are consistent because every kernel in (2.1) has mass one. This is the usual kernel extension

theorem [2]; conditioning on success would define a different experiment unless its preparation and waiting costs were separately charged.

For a fixed Bayesian preparation, let  $\pi$  be a reachable belief state on the hidden state, including the parameter when it is random. The positive instrument associated with an action and observation event is

$$\mathsf{T}_{a,B}\pi(E) = \int K(E, B, dc \mid x, a) \pi(dx).$$

After an observed event of positive probability, normalization gives the updated belief. For nonatomic reports we use a specified disintegration, with all equalities interpreted outside the appropriate null sets. A formula for an event of probability zero is not an additional observation channel.

**2.2. The future-test quotient.** A future test consists of a finite observation-based policy, a stopping rule, and a bounded measurable terminal payoff. Its conditional expectation at a belief  $\pi$  is written  $\pi(v)$ , where  $v$  is the corresponding backward test function. Fix a family  $\mathcal{V}$  containing constants and closed under the one-step pullbacks of the allowed tests. It may, for example, contain all bounded finite-horizon executable tests. Define

$$\pi \sim_{\mathcal{V}} \pi' \iff \pi(v) = \pi'(v) \quad (v \in \mathcal{V}). \quad (2.2)$$

The set of classes is the predictive quotient. The choice of tests is part of the experiment specification: identifying a hidden state is not required if none of the designated tests distinguishes it.

**Proposition 2.1** (Minimality and descent). *The quotient (2.2) is the coarsest set-valued statistic that determines all designated predictions. If the test family is one-step closed, normalized Bayesian updates descend to the quotient where their normalization is positive. If, in addition, the reachable belief domain  $B$  is compact metrizable, a countable family of continuous tests determines (2.2), and each fixed-report update is continuous on its positive-probability domain, the quotient has a canonical metrizable realization as a subset of  $\mathbb{R}^{\mathbb{N}}$ , and each such update descends continuously on compact subdomains on which its denominator is bounded away from zero.*

*Proof.* Map  $\pi$  to the vector of all its test expectations. Its fibres are exactly (2.2). Any statistic determining the expectations has fibres contained in these fibres, so the vector factors through that statistic as a set map. For the update, the numerator of a future prediction is a one-step pulled-back test and the denominator is the pulled-back constant. Equivalent states have equal numerators and denominators, hence equal normalized predictions.

In the countable continuous case use  $q(\pi) = (\pi(v_j))_{j \geq 1}$ . After scaling the tests to be bounded by one, its range is a compact subset of  $[-1, 1]^{\mathbb{N}}$ , with its product metric. A continuous map from a compact space into a Hausdorff space is a quotient map. The fibre-respecting continuous update therefore factors continuously on the stated compact subdomain. No assertion of a Borel cross-section, or of a smooth quotient for arbitrary Borel tests, is used. The minimality statement in the first paragraph alone would not imply either assertion.  $\square$

The proposition is relative to the prepared Bayesian experiment. An identity for one prior is not a parameter-uniform comparison of the family  $\{K^\lambda\}_\lambda$ . All experiment distances below explicitly take the supremum over the parameter when that stronger meaning is intended.

**2.3. Attainable Bayes geometry.** Let  $U$  be square integrable in a separable Hilbert space  $H$ , let  $\mathcal{H}$  be the actual information, and put  $m = \mathbb{E}[U \mid \mathcal{H}]$ . Define

$$e_M^2(m) = \inf_{z_1, \dots, z_M \in H} \mathbb{E} \min_i \|m - z_i\|^2.$$

Independent encoder/decoder randomization is permitted, but a retained observation is not a free input after encoding.

**Proposition 2.2** (Checkpoint geometry and information loss). *The optimum squared loss of a single  $M$ -label checkpoint statistic is*

$$\mathbb{E}\|U - m\|^2 + e_M^2(m). \quad (2.3)$$

*For any sub- $\sigma$ -field  $\mathcal{C} \subset \mathcal{H}$ , posterior laws of  $\lambda$  satisfy the barycentre identity  $\pi_{\mathcal{C}} = \mathbb{E}[\pi_{\mathcal{H}} \mid \mathcal{C}]$ . Consequently convex posterior functionals decrease under coarsening. Whenever the quantities are finite,*

$$I(\lambda; \mathcal{H}) - I(\lambda; \mathcal{C}) = \mathbb{E}D_{\text{KL}}(\pi_{\mathcal{H}} \parallel \pi_{\mathcal{C}}). \quad (2.4)$$

*In a differentiable dominated family, the score of a statistic  $S$  is  $\mathbb{E}[s_{\lambda} \mid S]$ , and the lost Fisher information matrix is  $\mathbb{E}\text{Cov}(s_{\lambda} \mid S)$ , provided differentiation and conditioning are justified in  $L^2$ .*

*Proof.* Conditional orthogonality gives  $\mathbb{E}\|U - z\|^2 = \mathbb{E}\|U - m\|^2 + \mathbb{E}\|m - z\|^2$  for every  $\mathcal{H}$ -measurable output, and also after adding independent randomness. Freeze that randomness. There are at most  $M$  decoder outputs, so nearest-centre assignment is no worse than the given encoding. Conversely every finite codebook has a measurable nearest assignment, with deterministic tie-breaking. Taking infima proves (2.3), without assuming attainment of an optimal codebook. Conditional expectation against bounded parameter tests gives the barycentre identity. Conditional Jensen proves the convex assertion. The chain rule for the logarithm of posterior densities gives (2.4); finite partitions followed by monotone approximation give the same assertion under the stated finiteness condition. Differentiating the marginal density of  $S$  gives the score identity. The matrix law of total variance gives the Fisher identity. These are the classical decision and information identities in [1, 11]; their role here is to specify the geometry being implemented, not to certify recursive closure.  $\square$

**2.4. Normalized response and observation modules.** Let  $b_{\theta}$  be a differentiable finite positive measure with  $b_{\theta}(1) > 0$ , and let  $\pi_{\theta} = b_{\theta}/b_{\theta}(1)$ . Differentiating this actual normalization gives

$$\partial_{\theta}\pi_{\theta}(f) = \frac{\partial_{\theta}b_{\theta}(f) - \pi_{\theta}(f)\partial_{\theta}b_{\theta}(1)}{b_{\theta}(1)}. \quad (2.5)$$

If the derivative has score density  $s$  relative to  $b_{\theta}$ , the right side is  $\text{Cov}_{\pi_{\theta}}(f, s)$ . Thus a response direction is invisible exactly when its centered score annihilates the linear span of the designated tests. If the observation protocol consists of products from two declared linear test spaces  $V, W$ , its null space is  $(VW)^{\perp}$ , with  $VW = \text{span}\{vw : v \in V, w \in W\}$ . This last statement follows immediately from linearity, but it requires that those products really are the observed probes. It does not identify an algebraic multiplicity with a physical noise variance. Formula (2.5) and this annihilator are the normalization and observation-module inputs used to type a contact experiment.

### 3. CAUSAL MORPHISMS AND CHARGED APPROXIMATION

A resource signature specifies ordered coordinates, for example persistent cardinality, numerical precision, transient workspace, program description, preparation count and physical time. A budget transformer is a monotone map between two such signatures, accompanied by an actual implementation. A coordinate may explicitly be unlimited. The persistent-cardinality results below leave fixed programs, fixed real constants, time and transient computation unlimited unless a more specific implementation is given. There is no claim that these resources are equivalent.

A causal simulation from an experiment  $\mathcal{E}$  to  $\mathcal{F}$  consists of a source controller, a causal transducer and its initialized state. For every target policy the joint construction must produce the target observation/cost history within a declared parameter-uniform total-variation error. The transducer never sees the unknown parameter or the unobserved physical state. Resource bounds are imposed on this joint construction, not just on the terminal observation kernel. We use  $\|P - Q\|_{\text{TV}} = \sup_A |P(A) - Q(A)|$ .

**Theorem 3.1** (Typed simulation and recursive approximation). *The following statements hold.*

- (i) *Serial causal simulations of errors  $\epsilon_1, \epsilon_2$  and budget transformers  $\rho_1, \rho_2$  compose with error at most  $\epsilon_1 + \epsilon_2$  and transformer  $\rho_1 \circ \rho_2$ . A simulator with  $K$  persistent states and a target observer with  $M$  states can be run with  $KM$  persistent states. For losses in  $[0, L]$ , the risk changes by at most  $L(\epsilon_1 + \epsilon_2)$ . Other resource coordinates are transformed according to the supplied implementations. The identity simulation has zero error and the identity transformer. Independent parallel simulations have additive TV error and the product register, with preparations and other costs charged by their joint implementation.*
- (ii) *Suppose the true predictive state lies in a metric space  $(S, d)$  and updates by  $s_t = F_t(s_{t-1}, y_t, a_t)$ . For all reports and actions suppose  $F_t$  is  $\rho$ -Lipschitz in its state,  $\rho < 1$ . Let an implementable approximate update  $\tilde{F}_t$  have local error at most  $\eta_t$ , and let  $Q_t : S \rightarrow C_t$  be a measurable finite quantizer with radius  $\epsilon_t$ . The register*

$$\hat{s}_t = Q_t \tilde{F}_t(\hat{s}_{t-1}, y_t, a_t) \quad (3.1)$$

*has at most  $|C_t|$  states at time  $t$  and, on every common observation path,*

$$d(s_t, \hat{s}_t) \leq \rho^t d(s_0, \hat{s}_0) + \sum_{j=1}^t \rho^{t-j} (\epsilon_j + \eta_j) =: b_t. \quad (3.2)$$

*A data-dependent persistent numerical residue must be part of  $C_t$ ; it cannot be placed in an uncharged side register. If the report kernels are  $L_0$ -Lipschitz in total variation as functions of the predictive state, the length- $n$  simulated history differs by at most  $L_0 \sum_{t < n} \mathbb{E} b_t$ . The same statement holds up to a bounded stopping time, with the stopped sum. If the true conditional mean is  $g(s_t)$  with  $g$   $L_1$ -Lipschitz into a Hilbert space, its implemented prediction has Bayes excess at most  $L_1^2 \mathbb{E} b_t^2$ .*

*Proof.* Compose the two transducers on their joint initialized state. Applying a Markov kernel contracts total variation, so the triangle inequality gives the sum of errors, uniformly over the policy and parameter for which the two certificates were supplied. The joint register is the Cartesian product. This proves the cardinality bound, and integration of a loss with range  $[0, L]$  proves the risk bound. The identity is the unchanged history. For independent parallel laws, insert one product factor at a time and use TV contraction to obtain the same sum bound. The composition order of the budget maps follows from first realizing the target in the intermediate experiment and then realizing that implementation in the source.

For (ii), insert  $F_t(\hat{s}_{t-1}, y_t, a_t)$  between the two updates. Contraction, the local defect and the quantization radius give  $d(s_t, \hat{s}_t) \leq \rho d(s_{t-1}, \hat{s}_{t-1}) + \eta_t + \epsilon_t$ . Induction proves (3.2). For the law comparison, telescope the products of the history kernels, replacing one kernel at a time. The prefix before the replaced kernel is distributed according to the true experiment; its integrated local TV error is at most  $L_0 \mathbb{E} b_t$ . All suffix kernels are contractions in TV. Summing proves the bound. An absorbing stopped state proves the stopped version by the identical calculation. The policy is evaluated on the same history in each local comparison: no continuity of a possibly discontinuous policy is presumed. Finally apply conditional orthogonality as in Proposition 2.2 and the Lipschitz bound for  $g$ .  $\square$

**Corollary 3.2** (Resource transport of an attainable profile). *If an  $M$ -state target observer has risk  $R_M$  and a  $K$ -state causal simulation has error  $\epsilon$  for its bounded decision problem, then the source has a  $KM$ -state observer of risk at most  $R_M + L\epsilon$ . Two simulations give the corresponding two directed inequalities; they need not preserve the same cardinality. A  $K$ -valued decoder-side event can always be absorbed into a register at a factor  $K$ , but a continuous side observation admits no such finite-factor assertion in general.*

*Proof.* Use the product construction. For the finite event store its current value as the second register coordinate. For the last assertion, an exact uniform real observation supplied directly to a decoder has zero squared estimation error with one otherwise empty state. If it is available only before an  $M$ -label encoding, scalar quantization has error  $1/(12M^2) > 0$ . Thus no finite factor simulates that side interface for all accuracies.  $\square$

The theorem joins quotient approximation to experiment comparison. Contractivity is a property to be verified on the true quotient, and  $Q_t$  is an executable representation of it. A finite-dimensional formula for the conditional mean is sufficient for one application, but is not a hypothesis of the theorem. In Section 6 the true state is an entire probability measure. In the next section a different structure replaces metric contraction by an exact finite-tree update.

#### 4. A STRUCTURAL THEOREM FOR PREDICTIVE TREES

**4.1. Local hypotheses and the finite profile.** Let  $\mathcal{L}$  be a pruned factorial language on a finite alphabet of size  $b \geq 2$ . Thus prefixes and suffixes of legal words are legal, every word has an infinite continuation, and only legal children occur in its rooted tree. Write  $[v]$  for a cylinder in the boundary  $\Omega$  and  $\emptyset$  for the empty word. Assume that along any branch there are at most  $U$  consecutive vertices with only one legal successor. This condition allows unary vertices but excludes an infinite deterministic branch. Let  $\mu$  be a stationary probability on  $\Omega$ , positive on every legal cylinder, and put  $p_v = \mu[v]$ .

A predictive energy is a positive family  $a_v$ , with  $a_\emptyset = 1$ . Put  $D_v = (a_v/p_v)^{1/2}$ . The following are the hypotheses of the structural theorem. There are constants  $a_- > 0$ ,  $\rho < 1$ , and  $c_0, C_0 > 0$  such that

$$a_- a_v \leq a_{vi} \leq \rho a_v, \quad \sum_{i: vi \in \mathcal{L}} a_{vi} \leq \rho a_v, \quad (4.1)$$

$$a_{uv} < a_v \quad (u \neq \emptyset), \quad D_{vi} \leq D_v. \quad (4.2)$$

These statements concern legal words only. A bounded Borel map  $Z$  from  $\Omega$  into a separable Hilbert space satisfies

$$\text{diam } Z([v]) \leq C_0 D_v, \quad (4.3)$$

$$\|Z(\omega) - Z(\omega')\| \geq c_0 D_u \quad \text{if } u \text{ is their longest common prefix.} \quad (4.4)$$

The image may be infinite dimensional. Separation is an explicit geometric hypothesis, not a consequence of the word notation.

Starting with the root, repeatedly split a leaf of maximal energy into *all* its legal children. Fix a tie rule. Let  $\mathcal{T}_N$  be the last tree in this sequence with at most  $N$  leaves,  $\mathcal{P}_N$  its leaf cut, and

$$G_N = \sum_{v \in \mathcal{P}_N} a_v. \quad (4.5)$$

A split at a unary vertex is an actual step. Finiteness of  $\mathcal{T}_N$  will follow below. The construction is a mathematical finite algorithm; computable implementation of arbitrary real energies is an additional input, not part of their Borel specification.

There are two causal interpretations of these data.

**Deletion model.** Independent acquisitions  $\omega \sim \mu$  occur at renewal times of independent positive integer length  $L$ , with  $\mathbb{E}L < \infty$ . The acquired sequence is observed exactly. At age  $k$  the desired output is  $g(\sigma^k \omega)$ , where  $g$  is bounded and Hilbert valued. Write

$$w_k = \mathbb{P}(L > k)/\mathbb{E}L, \quad W(n) = \sum_{k \geq n} w_k, \quad Z(\omega) = (\sqrt{w_k} g(\sigma^k \omega))_{k \geq 0}.$$

In addition to (4.1)–(4.4), assume the local oscillation bound

$$\sum_{k < |v|} w_k (\text{osc}_{[v_k:]} g)^2 + 4\|g\|_\infty^2 W(|v|) \leq C_1 D_v^2. \quad (4.6)$$

A register sees the renewal event only during its update; the decoder receives neither that event nor a stored raw observation for free.

**Insertion model.** Let  $(Y_t)_{t \in \mathbb{Z}}$  be a stationary finite-alphabet observed process whose reversed past  $\omega_t = (Y_t, Y_{t-1}, \dots)$  has law  $\mu$  and language  $\mathcal{L}$ . For a square-integrable stationary Hilbert target  $U_t$ , suppose

$$m_t = \mathbb{E}[U_t \mid Y_s : s \leq t] = Z(\omega_t), \quad B = \mathbb{E}\|U_t - m_t\|^2. \quad (4.7)$$

The finite register may start at zero with a fixed state; it is not initialized with the infinite past. Infinite stationary pasts here specify the reference prediction and its law. A finite warm-up is included in the limiting average. No renewal or independent-observation assumption is imposed in this interpretation.

For either model let  $R_M$  be the infimum, over all  $M$ -value registers with causal standard Borel update and decoding kernels, of the upper limiting expected average squared loss. Absolute time and independent coding randomization are permitted. In the insertion model put  $E_M = R_M - B$ .

**Theorem 4.1** (Predictive-tree realization in two orientations). *Under (4.1)–(4.4),*

$$cG_M \leq e_M^2(Z) \leq CG_M. \quad (4.8)$$

*Every complete greedy tree is suffix closed. A tree with  $L$  leaves has exactly  $1 + \sum_{v \text{ internal}} d^+(v)$  vertices, and at most  $(U+1)(2L-1)$  vertices; all of them may be used as designated states. For every fixed  $B_0 \geq 1$ ,  $G_{\lceil B_0 M \rceil} \asymp_{B_0} G_M$ .*

*In the deletion model, with (4.6),*

$$cG_M \leq e_M^2(Z) \leq R_M \leq CG_M. \quad (4.9)$$

*In the insertion model,*

$$cG_M \leq e_M^2(Z) \leq E_M \leq CG_M. \quad (4.10)$$

*The converses apply to the lower limiting average of each randomized, time-dependent observer. The upper bounds are achieved in order by stationary deterministic updates. In particular a logarithmic age or history multiplier is not required. Constants depend on the displayed structural hypotheses, not on the register budget.*

## 4.2. Balance, dilation and exact closure.

**Lemma 4.2** (Greedy balance). *For all sufficiently large  $N$ ,  $N/b \leq |\mathcal{P}_N| \leq N$ . If  $\theta_N$  is the largest energy of a current leaf, then*

$$a_- \theta_N \leq a_v \leq \theta_N \quad (v \in \mathcal{P}_N). \quad (4.11)$$

*The trees are finite and their profiles have the fixed-dilation property in Theorem 4.1.*

*Proof.* Suppress unary vertices in any finite complete tree. A rooted tree with  $L$  leaves and no unary internal vertex has at most  $2L - 1$  vertices, since its number of edges is the sum of its internal outdegrees. Each vertex of the suppressed tree accounts for itself and at most  $U$  unary vertices on its incoming path, including the initial root path. Thus the original tree has at most  $(U+1)(2L-1)$  vertices. An infinite sequence of splits with bounded leaf number is impossible, proving finiteness. The split following  $\mathcal{T}_N$  crosses the budget and adds at most  $b-1$  leaves. Hence  $|\mathcal{P}_N| > N - b + 1$ .

The maximal leaf energy never increases. Every present leaf was born with energy at least  $a_-$  times an earlier maximal energy. This proves (4.11), including any unary splits. Set  $\ell = U + 1$ . Every vertex has at least two distinct descendants at depth  $\ell$ , and at least  $2^j$  at depth  $\ell j$ . Choose



$j$  with  $2^j > b(B_0 + 2)$ . If the later maximal energy at budget  $\lceil B_0 N \rceil$  were less than  $a_-^{\ell_j+1} \theta_N$ , every original leaf would have been split through all these  $\ell_j$  levels: all intervening vertices have energy at least that threshold. The resulting number of leaves would exceed  $\lceil B_0 N \rceil$ , a contradiction. Therefore the later maximum is at least a fixed multiple of  $\theta_N$ . Applying (4.11) at the later budget gives  $G_{\lceil B_0 N \rceil} \geq c_{B_0} N \theta_N \geq c_{B_0} G_N$ . Splitting decreases the energy sum by (4.1), giving the reverse inequality. Adjusting constants covers the finite initial range.  $\square$

**Lemma 4.3** (Two exact operations on one tree). *Every proper suffix of an internal greedy-tree vertex is internal, and every suffix of any vertex is present. Consequently deletion of the first letter stays in the tree. If  $v$  is a leaf and  $iv$  is legal, then  $iv$  has a leaf prefix in the same tree. It can therefore be updated by prepending  $i$  and reading only the letters of  $v$  until that leaf is reached.*

*Proof.* Suppose an internal word  $uv$ ,  $u \neq \emptyset$ , is about to be split. If its suffix  $v$  has not been split, some current leaf  $t$  is a prefix of  $v$ . Child decrease and strict prefix-attachment decrease give  $a_t \geq a_v > a_{uv}$ , contrary to maximality. Thus the suffix is already internal. For a leaf  $i_1 \cdots i_n$  with  $n \geq 2$ , its parent is internal; the suffix  $i_2 \cdots i_{n-1}$  of that parent is internal, so its legal child  $i_2 \cdots i_n$  is present. The one-letter case ends at the root. Iteration gives closure of every vertex.

If there were no leaf prefix of  $iv$ , the complete tree would contain  $iv$  as an internal vertex. Its suffix  $v$  would then be internal by the first assertion, contradicting that it is a leaf. This proves the insertion assertion. Only legal children were used in either proof.  $\square$

#### 4.3. Static lower bound and the causal converse.

**Lemma 4.4** (Arbitrary centres). *The bounds (4.8) hold for arbitrary Hilbert-space centres, not only centres on  $Z(\Omega)$ .*

*Proof.* Choose one image point in each leaf cylinder to obtain  $e_N^2(Z) \leq C_0^2 G_N$ . For the lower bound use  $\mathcal{P}_{4bM}$ , which has at least  $4M$  leaves for large  $M$ . The open tubes about  $Z([v])$  of radii  $\delta D_v$  are disjoint for a sufficiently small fixed  $\delta > 0$ . Indeed for incomparable  $v, v'$  their common prefix  $u$  has  $D_u \geq \max(D_v, D_{v'})$ , and (4.4) separates the two images by at least  $c_0 D_u$ . Take  $2\delta < c_0$ .

At most  $M$  of the tubes contain one of  $M$  proposed centres. On every unoccupied cylinder the squared error is at least  $\delta^2 D_v^2$ . Its integrated contribution is  $\delta^2 a_v$ . By balance, the at least three-quarters unoccupied leaves carry at least  $3a_-/4$  of the total energy sum. Thus  $e_M^2(Z) \geq cG_{4bM} \geq c'G_M$  by Lemma 4.2. The finite initial range is harmless: (4.4) and positive cylinder masses at unbounded branching depths give infinitely supported image law, so every fixed finite-centre error is positive. Alternatively the same tube proof at a larger fixed cut proves that positivity directly.  $\square$

**Lemma 4.5** (Renewal converse). *In the deletion experiment every randomized, nonstationary  $M$ -state observer has lower limiting average loss at least  $e_M^2(Z)$ .*

*Proof.* A standard Borel probability kernel is realized as a measurable function of its input and an independent uniform random variable [2]. Applying this at the countably many update and decode times gives an independent coding sequence. A data-dependent persistent position in such a sequence, if used, belongs to the charged state; the representation itself supplies no history channel.

Condition on an acquisition at a fixed time, its pre-acquisition history, and the independent coding variables. The acquired  $\omega$  still has law  $\mu$ . Along a hypothetical continuation with no new acquisition, each of the at most  $M$  post-acquisition labels determines one output string. For a truncation at age  $n$ , the survival-weighted loss is therefore at least  $\mathbb{E} L e_M^2(Z^{<n})$ , where  $Z^{<n}$  is the first  $n$  coordinates of the weighted orbit. At a fixed physical time, events specifying the last acquisition are disjoint. Summing this bound over acquisition times at least  $n$  before the end of a

horizon and using the Cesàro renewal count gives lower limiting average at least  $e_M^2(Z^{<n})$ . For completeness, the elementary renewal theorem gives expected renewals per unit time tending to  $1/\mathbb{E}L$ ; a fixed truncation removes at most  $n$  terminal time indices, each with bounded contribution. No stationary coding rule is presumed.

Finally  $e_M^2(Z^{<n}) \uparrow e_M^2(Z)$ . Monotonicity follows by projection. The reverse limiting inequality follows by appending zero tails to an arbitrarily good truncated code: the extra mean squared norm is at most  $\|g\|_\infty^2 W(n) \rightarrow 0$ . This completes the proof.  $\square$

*Proof of Theorem 4.1.* The static assertion, finite count, dilation and suffix closure are proved above. In the deletion experiment retain every tree vertex as a designated state. At acquisition store the unique leaf prefix of  $\omega$ . Between acquisitions delete its first letter until the empty word is reached, and then keep that word. Choose a representative in each nonempty cylinder and output its  $g$  value; use a fixed bounded output at the empty word. For an acquired leaf  $v$  the weighted error is bounded by the left side of (4.6). Averaging and applying renewal reward gives at most  $C_1 G_N$ . Lemma 4.5 gives the unrestricted converse.

For insertion, after reaching a leaf use the prepend-and-parse rule of Lemma 4.3. During initialization store the observed reversed word while it is internal, or its first leaf prefix when one appears. This uses only vertices of the same tree and after at most its maximum depth gives the correct leaf prefix of the full observed past. Decode a representative  $Z$  value in that leaf cylinder. Its stationary excess loss is at most  $C_0^2 G_N$  by (4.3) and conditional orthogonality. The finite warm-up has zero limiting average contribution. The converse holds at each time: freeze independent coding randomness, apply orthogonality against  $m_t$ , and use the at most  $M$  decoder values to lower bound excess by  $e_M^2(Z)$ . Stationarity makes this the same number at every time, even for time-dependent decoders.

In either orientation choose  $N$  a fixed sufficiently small multiple of  $M$ , so that  $(U+1)(2|\mathcal{P}_N|-1) \leq M$ . Dilation converts  $G_N$  into  $O(G_M)$  without a time-depth multiplier. One fixed state handles the remaining bounded range of budgets. This proves the two risk assertions.  $\square$

The static part of the proof uses familiar separated-cylinder quantization ideas [9, 6]. Maximal-energy prefix splitting is likewise a classical coding device [7]. The additional conclusion is that a strict *predictive* energy inequality makes the adaptive tree executable in both orientations, with its whole persistent state counted.

## 5. TWO MODEL VERIFICATIONS

**5.1. Markov-renewal orbit geometry.** Let  $A$  be a primitive zero-one matrix on  $b$  symbols, with Perron root greater than one. Its legal one-sided sequences have a stationary fully supported law  $\mu$  such that every legal conditional child probability is at least  $\zeta > 0$ . Let  $h_i$  map a compact set of diameter at most one into pairwise  $g$ -separated subsets of itself. Assume word scales  $r_v$ ,  $r_\emptyset = 1$ , satisfy

$$\begin{aligned} D_0^{-1} r_v |x - y| &\leq |h_v x - h_v y| \leq r_v |x - y|, \\ D_0^{-1} r_u r_v &\leq r_{uv} \leq r_u r_v, \quad 0 < r_- \leq r_i \leq r_+ < 1. \end{aligned} \quad (5.1)$$

The injective coding map identifies the legal sequences with a compact set  $K$ , on which  $T$  deletes the first symbol. Let  $f : K \rightarrow \mathbb{R}^m$  obey

$$|f(x) - f(y)| \leq H|x - y|^\beta, \quad \sum_{j=0}^J |f(T^j x) - f(T^j y)|^2 \geq c_o |x - y|^{2\beta}, \quad 0 < \beta \leq 1. \quad (5.2)$$

The renewal age weights satisfy  $0 < q_- \leq w_{k+1}/w_k \leq q_+ < 1$ . Set  $\tau = 2\beta$  and

$$V(v) = \sum_{k < |v|} w_k r_{v_k}^\tau + W(|v|), \quad a_v = \mu[v]V(v), \quad Z(x) = (\sqrt{w_k} f(T^k x))_{k \geq 0}. \quad (5.3)$$

**Corollary 5.1** (Markov renewal as a structural instance). *The data (5.1)–(5.3) satisfy Theorem 4.1 in its deletion orientation. In particular  $e_M^2(Z) \asymp R_M \asymp G_M$ , with a stationary finite-tree observer and an unrestricted randomized, nonstationary converse. Forbidden edges are not completed, and unary vertices are charged.*

*Proof.* Every suffix scale is at most one and appending a letter decreases its scale. If  $|u| = m$  and  $|v| = n$ , monotonicity of  $w$  gives

$$\begin{aligned} V(v) - \left( \sum_{j < n} w_{m+j} r_{v_j}^\tau + W(m+n) \right) &\geq r_v^\tau \sum_{k < m} w_k, \\ V(uv) - \left( \sum_{j < n} w_{m+j} r_{v_j}^\tau + W(m+n) \right) &\leq r_+^\tau r_v^\tau \sum_{k < m} w_k. \end{aligned}$$

Therefore  $V(uv) < V(v)$ . Stationarity gives only  $\mu[uv] \leq \mu[v]$ , which is enough to give the strict energy inequality. Equality of cylinder masses at a unique predecessor causes no problem.

Appending a legal letter multiplies old spatial terms by a factor between  $D_0^{-\tau} r_-^\tau$  and  $r_+^\tau$ . The old tail  $W(n)$  becomes  $w_n r_i^\tau + W(n+1)$ . Since  $w_n \geq (1 - q_+)W(n)$  and  $W(n+1) \geq q_- W(n)$ , we have

$$\ell V(v) \leq V(vi) \leq \rho V(v), \quad \ell = \min(D_0^{-\tau} r_-^\tau, q_-), \quad \rho = 1 - (1 - r_+^\tau)(1 - q_+) < 1.$$

Use  $a_- = \zeta \ell$  and sum the conditional masses to verify (4.1). These inequalities also give monotonicity of  $D_v = \sqrt{V(v)}$ . In a primitive graph with Perron root greater than one, a deterministic path cannot run for  $b$  repeated unary vertices: a repeated vertex would generate a closed deterministic cycle. Thus one can take  $U = b$ .

Introduce  $d_\beta(x, y)^2 = \sum_k w_k |T^k x - T^k y|^{2\beta}$ . Hölder continuity bounds the orbit distance from above by  $H d_\beta$ . Summing the delayed lower bound (5.2), and using  $w_{k-j} \leq q_-^{-j} w_k$ , bounds  $d_\beta$  from above by a constant times the orbit distance. Points in one cylinder have squared  $d_\beta$  distance at most  $V(v)$ . Points whose longest common prefix is  $u$  have, before its end, distances at least  $D_0^{-1} g r_{u_k}$ , and at its end distance at least  $g$ . The last term controls  $W(|u|)$  since  $W(n) \leq w_n/(1 - q_+)$ . This proves both image comparisons. Finally the Hölder bound on each suffix and the bounded tail prove (4.6). The structural theorem now supplies all combinatorial, quantization, and causal conclusions.  $\square$

This corollary recovers the earlier Markov-renewal profile without repeating its tree argument as a new model-specific proof. Gibbs pressure identifies the profile exponent under the additional thermodynamic assumptions of the previous Markov theorem, whose full proof is retained in the companion. Pressure is a computation of this instance, not an assumption of the foundational realization theorem. For clarity, the passage from power bounds to an exponent uses only the following elementary fact.

**Proposition 5.2** (Logarithmic squeeze). *If for every  $\delta > 0$  a positive sequence  $R_M$  satisfies  $c_\delta M^{-\alpha-\delta} \leq R_M \leq C_\delta M^{-\alpha+\delta}$  eventually, then  $-\log R_M / \log M \rightarrow \alpha$ .*

*Proof.* Taking logarithms gives lower and upper limits between  $\alpha - \delta$  and  $\alpha + \delta$ , because fixed logarithmic constants divided by  $\log M$  vanish. Let  $\delta$  decrease to zero. In a pressure-root proof one first chooses the auxiliary root parameter on either side of its zero and only then lets  $M$  tend to infinity; root continuity then supplies the arbitrary  $\delta$  used here.  $\square$

**5.2. An infinite hidden shift with noisy continual reports.** Let  $(B_t)_{t \in \mathbb{Z}}$  be independent identically distributed Bernoulli variables. The hidden state at time  $t$  is the whole sequence  $X_t = (B_t, B_{t-1}, \dots)$ . A binary memoryless channel reports  $Y_t$  about  $B_t$ . Write

$$\gamma = \mathbb{P}(Y_t = 1) \in (0, 1), \quad a_i = \mathbb{P}(B_t = 1 \mid Y_t = i), \quad a_0 \neq a_1.$$

Fix  $0 < r < 1$ . The Hilbert-valued prediction target and its mean are

$$U_t = (\sqrt{1 - r^2} r^j B_{t-j})_{j \geq 0}, \quad m_t = (\sqrt{1 - r^2} r^j a_{Y_{t-j}})_{j \geq 0}. \quad (5.4)$$

This squared loss can equally be implemented by drawing a fresh terminal query  $j$  with probability  $(1 - r^2)r^{2j}$  and scoring the prediction of  $B_{t-j}$ . The independent query is a task index, not a new observation of the hidden state. Its distribution and interface are fixed in advance.

**Theorem 5.3** (Sharp register law for an infinite posterior). *Put  $p_0 = 1 - \gamma$ ,  $p_1 = \gamma$  and let  $s \in (0, 1)$  be the unique solution of*

$$r^{2s}(p_0^s + p_1^s) = 1, \quad \alpha = (1 - s)/s. \quad (5.5)$$

*For the actual noisy shift experiment,*

$$R_M = B + \Theta((a_1 - a_0)^2 M^{-\alpha}), \quad B = \sum_{i=0}^1 p_i a_i (1 - a_i). \quad (5.6)$$

*The constants in the excess comparison depend on  $r, p_0, p_1$ , not on  $a_0, a_1$ . No posterior truncation is assumed in deriving this law. For a fair hidden bit and a binary symmetric channel of error  $\varepsilon < 1/2$ , it becomes*

$$R_M = \varepsilon(1 - \varepsilon) + \Theta((1 - 2\varepsilon)^2 M^{2 \log r / \log 2}).$$

*At  $\varepsilon = 1/2$  the predictive mean is constant and the excess is zero with one state.*

*Proof.* The full posterior is the product of Bernoulli laws with parameters  $a_{Y_{t-j}}$ . Thus (5.4) is the actual conditional mean, and orthogonality gives the displayed  $B$ . Normalize its nonconstant part by  $a_1 - a_0$ . On the full binary past language take  $p_v = \prod p_{v_j}$ ,  $D_v = r^{|v|}$  and  $a_v = p_v r^{2|v|}$ . Children multiply energy by  $p_i r^2$ , their sum multiplies it by  $r^2$ , and every nonempty prefix attachment strictly decreases it. If two pasts first differ at position  $n$ , their normalized squared image distance lies between  $(1 - r^2)r^{2n}$  and  $r^{2n}$ . All hypotheses of Theorem 4.1 hold in the insertion orientation.

For the exact order of  $G_M$ , set  $q_i = (p_i r^2)^s$ . Equation (5.5) says  $q_0 + q_1 = 1$ . For every finite complete cut, consistency of this Bernoulli measure gives  $\sum_v a_v^s = 1$ . By greedy balance, for a cut with  $L$  leaves and maximum  $\theta$ ,  $L(a_- \theta)^s \leq 1 \leq L\theta^s$ . Hence  $G_M \asymp L\theta \asymp L^{-(1-s)/s} \asymp M^{-\alpha}$ . The root exists uniquely because the left side of (5.5) decreases continuously from two to  $r^2 < 1$ . The fair case solves it explicitly, and the constant-mean case is immediate. If only reports from time zero are supplied, the unobserved past contributes at most  $r^{2(t+1)}$  to the squared target norm at time  $t$ . Its average vanishes; the finite-tree initialization in the structural theorem still gives (5.6).  $\square$

This is not an exactly closed finite list of conditional means. Every past report changes its own posterior coordinate, and arbitrarily many coordinates can be queried. It also has no fixed-step common minorizer: after  $k$  hidden updates the old tail remains intact, and two different initial tails give disjoint supports. A measure dominated by all those kernels must be zero. As always, a bare Borel encoding of an infinite sequence into one real number is possible; it is not a finite-cardinality register and does not change the theorem.

## 6. FINITE PRESENTATIONS OF MEASURE-VALUED FILTERS

The preceding sharp example has a separated tree geometry. We next give a constructive approximation theorem requiring no such separation and no finite-dimensional exact posterior coordinate.

**Lemma 6.1** (A finite Wasserstein presentation). *For  $n \geq 1$  let  $\mathcal{C}_n$  consist of empirical probability measures with  $n$  equally weighted atoms on  $\{0, 1/n, \dots, 1\}$ . Then*

$$|\mathcal{C}_n| = \binom{2n}{n} \leq 4^n, \quad \sup_{\pi \in \mathcal{P}([0,1])} W_1(\pi, Q_n \pi) \leq \frac{3}{2n} \quad (6.1)$$

for a specified Borel quantizer  $Q_n$ .

*Proof.* Take the generalized quantiles of  $\pi$  at  $(j - 1/2)/n$ ,  $1 \leq j \leq n$ , and round them to nearest grid points, breaking ties toward zero. Quantiles are Borel functions of a probability law. If  $q$  is its monotone quantile function, the integral of the error from replacing  $q$  by its midpoint value on each of the  $n$  equal subintervals is at most  $1/n$ : bound each integral by its interval length times the oscillation and sum the monotone variations, at most one. Rounding adds at most  $1/(2n)$ . In one dimension  $W_1$  is the  $L^1$  distance of quantile functions, giving the bound. The possible sorted lists are multisets of size  $n$  from  $n + 1$  points, counted by  $\binom{2n}{n}$ . The binomial theorem bounds this by  $4^n$ .  $\square$

**Theorem 6.2** (A charged measure-valued recursive approximation). *Let the actual filter obey  $\pi_t = \Phi_{y_t, a_t}(\pi_{t-1})$  on  $\mathcal{P}([0, 1])$ , and suppose every allowed update is  $\rho$ -contractive in  $W_1$ , with  $\rho < 1$ . Suppose an implementable update has  $W_1$  defect at most  $\eta$ . The initialized register*

$$\hat{\pi}_t = Q_n \tilde{\Phi}_{y_t, a_t}(\hat{\pi}_{t-1}), \quad \hat{\pi}_0 \in \mathcal{C}_n$$

has at most  $\binom{2n}{n}$  persistent values and satisfies, for every actual observation path,

$$W_1(\pi_t, \hat{\pi}_t) \leq \rho^t + \frac{3/(2n) + \eta}{1 - \rho}. \quad (6.2)$$

This bounds the error of every 1-Lipschitz test expectation, uniformly over the choice of test. Thus for  $M \geq 4$ ,  $n = \lfloor \log M / \log 4 \rfloor$  gives error  $O((\log M)^{-1} + \eta)$  and squared error  $O((\log M)^{-2} + \eta^2)$  after the initialization term. If these test expectations are conditional means, the latter is a Bayes excess bound. Report-law comparison additionally requires the report Lipschitz hypothesis of Theorem 3.1; it is not automatic from  $W_1$  contraction.

*Proof.* Apply Theorem 3.1 with the true metric state  $S = \mathcal{P}([0, 1])$ , radius  $3/(2n)$ , and diameter one. The dual formula for  $W_1$  gives the assertion for every test; independent selection of a terminal test does not enlarge its bound. The codebook count and the logarithmic substitution are exact. Only an index of a codebook law persists between updates, not its exact unquantized posterior or a numerical residual.  $\square$

**6.1. A nonlinear physical realization.** Let  $B_t$  be independent fair bits and define

$$h_0(x) = x/5 + x^2/50, \quad h_1(x) = 3/5 + x/5 + x^2/50, \quad X_t = h_{B_t}(X_{t-1}). \quad (6.3)$$

The observation  $Y_t$  reports  $B_t$  correctly with probability  $3/4$ , independently at each time. A permitted terminal probe  $g : [0, 1] \rightarrow [0, 1]$  returns a Bernoulli variable of probability  $g(X_t)$ ; the tested family is 1-Lipschitz probes and their bounded finite executable continuations. The actual update-report instrument on measures is

$$\mathsf{T}_y \pi = \frac{1}{2} \left\{ \frac{3}{4} (h_y)_\# \pi + \frac{1}{4} (h_{1-y})_\# \pi \right\}.$$

Its mass is  $1/2$ , independent of the old state. The normalized filter is therefore

$$\Phi_y \pi = \frac{3}{4} (h_y)_\# \pi + \frac{1}{4} (h_{1-y})_\# \pi. \quad (6.4)$$

**Corollary 6.3** (Nonlinear posterior, explicit register). *The actual measure-valued filter (6.4) is 6/25-contractive in  $W_1$ . Theorem 6.2 applies with  $\eta = 0$  to an exact rational implementation on  $C_n$ . The model has no fixed-step nonzero common minorizer. For every integer  $D \geq 1$ , the first  $D$  polynomial moments do not determine their own exact next-step update on the space of all prior laws.*

*Proof.* The two images lie respectively in  $[0, 11/50]$  and  $[3/5, 41/50]$ . Each derivative is between  $1/5$  and  $6/25$ . Couple two old laws optimally and use the same branch draw in their mixtures; this gives  $W_1(\Phi_y \pi, \Phi_y \pi') \leq (6/25)W_1(\pi, \pi')$ . For a codebook input, (6.4) has at most  $2n$  atoms, at rational points with denominator dividing  $50n^2$ , with weights multiples of  $1/(4n)$ . Sorted cumulative weights give the exact quantiles, which are rounded back to the fixed grid. Thus no numerical limit is being hidden in  $\eta = 0$ . The transient list of at most  $2n$  atoms is discarded; its entries and weights have  $O(\log n)$ -bit rational descriptions. The designated persistent object is the single codebook index. Approximate arithmetic instead obeys the separately charged  $\eta$  bound of the theorem.

After  $k$  hidden transitions the law from a fixed  $x$  has finite support  $\{h_v(x) : |v| = k\}$ . For a second initial point  $x'$ , each equality  $h_v(x') = h_u(x)$  excludes at most one  $x'$ , since  $h_v$  is strictly increasing. Choose  $x'$  outside this finite set. The two supports are disjoint, so a measure dominated by all  $k$ -step laws must be zero.

For the moment assertion, the  $D$ th updated moment integrates  $\frac{3}{4}h_y(x)^D + \frac{1}{4}h_{1-y}(x)^D$ , a polynomial of degree  $2D$  with nonzero leading coefficient  $50^{-D}$ . It is not in the span of  $1, x, \dots, x^D$ . Linear algebra on finitely many distinct points therefore supplies a signed measure annihilating this span but not the new polynomial. Its positive and negative parts have equal mass; normalizing them yields two prior probabilities with identical first  $D$  moments and different updated  $D$ th moment. The argument is about ordinary finite moment closure, not the impossibility of a Borel encoding of measures by real numbers.  $\square$

The quotient here is genuinely the posterior law for the specified probe family: bounded Lipschitz functions separate probability measures on a compact interval, and constants do not affect that separation. Updates respect this quotient by (6.4). The advancing reports have the exact state-independent fair law, while a terminal Bernoulli probe has total-variation defect at most  $W_1(\pi_t, \hat{\pi}_t)$ . Thus the same finite presentation defines a causal experiment simulation with a controlled terminal error and an exactly charged register. Multiple probes would update the posterior again; no claim for an unverified enlarged probe protocol is implicit. This example connects positive instruments, predictive quotient, recursive approximation and resource transformation within one model.

## 7. SHARP FINITE-EXPERIMENT BUDGETS

For experiments  $\mathcal{E} = \{P_\theta\}$  and  $\mathcal{F} = \{Q_\theta\}$  with the same parameter set, write

$$\delta(\mathcal{E}, \mathcal{F}) = \inf_K \sup_\theta \|K P_\theta - Q_\theta\|_{\text{TV}}, \quad \Delta(\mathcal{E}, \mathcal{F}) = \max\{\delta(\mathcal{E}, \mathcal{F}), \delta(\mathcal{F}, \mathcal{E})\}.$$

All parameterized laws in this section are Borel families. The infimum is over parameter-independent Markov kernels. These are terminal experiment comparisons in the sense of [11, 12]; only the encoding statistic, not the comparison randomization, is part of the physical data reduction.

Let  $\mathcal{G} = \{N(z, \Gamma) : z \in K\}$ , where  $K \subset \mathbb{R}^r$  is compact and  $\Gamma$  is positive definite. In the lower bound assume that  $K$  contains a nonempty Euclidean ball.

**Theorem 7.1** (Gaussian finite-experiment cardinality). *There are constants  $c, C > 0$ , depending on  $K, \Gamma, r$ , such that*

$$cM^{-2/r} \leq \inf_{\mathcal{F}: |\mathcal{F}| \leq M} \Delta(\mathcal{F}, \mathcal{G}) \leq CM^{-2/r}, \quad M \geq 1. \quad (7.1)$$

The lower bound ranges over all experiments on at most  $M$  outcomes, not just deterministic Gaussian quantizers. The upper bound is attained in order by a deterministic statistic with  $J^r$  labels, where  $J = \lfloor M^{1/r} \rfloor$ , followed by a specified positive reconstruction kernel. No overflow label is needed.

### 7.1. The cardinality obstruction.

**Lemma 7.2** (A decision-theoretic lower bound). *Under the preceding assumptions,  $\Delta(\mathcal{F}, \mathcal{G}) \geq cM^{-2/r}$  whenever  $\mathcal{F}$  has at most  $M$  outcomes.*

*Proof.* Restrict the mean parameter to a closed cube  $K_0$  of positive side length contained in  $K$  and put the uniform prior on  $K_0$ . Restriction can only decrease an experiment distance. Use squared-error estimation of  $z$ , with actions restricted to  $K_0$ ; its loss range is bounded by  $L = \text{diam}(K_0)^2$ . Let  $m(y) = \mathbb{E}[z \mid y]$  in this Gaussian experiment. Differentiation under the compact prior integral gives

$$Dm(y) = \text{Cov}(z \mid y)\Gamma^{-1}. \quad (7.2)$$

The posterior density is positive on the cube. Hence its covariance is positive definite, and  $Dm(y)$  is nonsingular for every  $y$ . The inverse function theorem on a small neighborhood of any fixed  $y_0$ , together with the positive continuous marginal density of  $y$ , shows that the law of  $m(y)$  has a density bounded below by a positive constant on some smaller cube in  $\mathbb{R}^r$ .

This implies  $e_M^2(m) \geq c_0 M^{-2/r}$ . To see it directly, put balls of radius  $t = (|Q|/(2Mv_r))^{1/r}$  about the  $M$  proposed centres, where  $Q$  is the smaller cube and  $v_r$  the volume of the unit ball. At least half of  $Q$  remains outside them, and contributes at least a fixed multiple of  $t^2$ . This argument permits arbitrary centres.

Every  $M$ -outcome garbling  $S$  of the Gaussian observation therefore has Bayes risk at least  $B_{\mathcal{G}} + c_0 M^{-2/r}$ , by conditional orthogonality. If  $\Delta(\mathcal{F}, \mathcal{G}) < \epsilon$ , choose a kernel  $K$  from  $\mathcal{G}$  to  $\mathcal{F}$  with error below  $\epsilon$ . The experiment  $K\mathcal{G}$  has at most  $M$  outputs, and bounded-loss comparison gives

$$B_{K\mathcal{G}} \leq B_{\mathcal{F}} + L\epsilon \leq B_{\mathcal{G}} + 2L\epsilon.$$

The second inequality uses a kernel in the reverse direction. Comparing these bounds and taking the infimum over  $\epsilon$  proves the result. Randomized actions do not improve a Bayes squared-error minimum. Notice that  $\mathcal{F}$  itself was not assumed to be a Gaussian garbling.  $\square$

### 7.2. Positive second-order reconstruction. Put

$$\tau(x) = \frac{1}{2} + \pi^{-1} \arctan x, \quad u = \tau^{\otimes r}(y) \in (0, 1)^r. \quad (7.3)$$

Let  $p_z$  be the transformed Gaussian density. If  $x_j = \tan \pi(u_j - 1/2)$ , then

$$p_z(u) = \phi_{\Gamma}(x - z) \prod_{j=1}^r \pi(1 + x_j^2). \quad (7.4)$$

It extends by zero across every face to a smooth periodic density. Indeed each fixed-order derivative is a polynomial in  $x$  times the Gaussian density, which tends to zero faster than any polynomial as a face is approached. Uniformity holds for  $z$  in the fixed compact set. For use below define the finite constants

$$H_2 = \sup_{z, u} \|D^2 p_z(u)\|_{\text{op}}, \quad H_0 = \max_j \sup_{z, u_j} p_{z, j}(u_j), \quad (7.5)$$

where  $p_{z, j}$  is a transformed marginal density. These suprema, or any upper bounds obtained from (7.4), fix all the reconstruction and boundary constants; they do not depend on  $M$  or sample size. For instance, differentiating twice introduces only polynomials and the eigenvalue bounds of  $\Gamma$ , so bounded mean norm and fixed covariance conditioning suffice.

Partition the flat unit torus into  $J^r$  cubes of side  $h = 1/J$ , with centres  $c_i$ , and write  $Q_J(u)$  for the cube label. For  $J \geq 2$  let  $\Phi_i$  be the periodic tensor product of the piecewise-linear hat

functions with value one at  $c_i$ , zero at its neighboring mesh centres, and support within one mesh step in each coordinate. Then

$$\Phi_i \geq 0, \quad \int \Phi_i = h^r, \quad \sum_i \Phi_i(u) = 1.$$

On each mesh interval the neighbouring centres reproduce the affine coordinate, using local periodic lifts across the seam. Define the reconstruction from label  $i$  to have density  $h^{-r}\Phi_i(u)$ , then apply  $\tau^{-1}$  coordinatewise. Values on the faces have probability zero.

**Lemma 7.3** (A quadratic reconstruction estimate). *Uniformly in  $z \in K$ , this kernel  $H_J$  satisfies*

$$\|H_J Q_J \tau^{\otimes r} N(z, \Gamma) - N(z, \Gamma)\|_{\text{TV}} \leq \frac{rH_2}{12J^2}. \quad (7.6)$$

*Proof.* Let  $p_i$  be the probability of cube  $i$ . Symmetric Taylor expansion about its centre gives  $|p_i/h^r - p_z(c_i)| \leq rH_2h^2/24$ . The linear interpolant  $I_h p_z = \sum_i p_z(c_i)\Phi_i$  has error at most  $rH_2h^2/8$ : its weights reproduce constants and affine functions and the sum of their coordinate variances is at most  $rh^2/4$ . The second Taylor remainder is half this variance times  $H_2$ . Both calculations hold across the seam because  $p_z$  has a smooth periodic extension. The reconstructed density is  $\sum_i (p_i/h^r)\Phi_i$ . Nonnegativity and the partition of unity give supremum error at most  $rH_2h^2/6$ . Integrating over volume one and using the factor  $1/2$  for total variation proves the claim. Inverting (7.3) preserves this distance.  $\square$

*Proof of Theorem 7.1.* Lemma 7.2 gives the lower bound. The deterministic statistic  $Q_J \tau^{\otimes r}$  has  $J^r$  outputs, so the forward deficiency is zero; Lemma 7.3 gives the reverse deficiency. For  $M \geq 2^r$ ,  $J = \lfloor M^{1/r} \rfloor \geq M^{1/r}/2$ . For the remaining finite range use any one-outcome experiment and the trivial bound  $\Delta \leq 1$ , increasing  $C$ . The reconstruction uses positive probabilities; it is not a signed interpolation operator.  $\square$

The use of a compactified grid avoids a growing Gaussian overflow radius. The use of hats rather than constant cells cancels the first Taylor term. Both choices are necessary for the stated proof of a logarithm-free quadratic budget rate.

## 8. A SHARP BUDGET FOR THE PHYSICAL COUNT-TO-CONTACT EXPERIMENT

We now specify the actual local protocol, following the known-mark experiment of A2 [8]. Fix  $\lambda_0 \in (0, \infty)^m$ , a matrix  $B \in \mathbb{R}^{m \times d}$ , a compact set  $H \subset \mathbb{R}^d$ , and a full-row-rank matrix  $C \in \mathbb{R}^{r \times m}$ . Observe independent counts

$$K_{N,j} \sim \text{Pois}(N\lambda_{0,j} + \sqrt{N}(Bh)_j), \quad h \in H. \quad (8.1)$$

Write  $\Lambda = \text{diag } \lambda_0$ ,  $A = CB$ , and  $\Gamma = C\Lambda C^T$ . The contact score and limiting experiment are

$$S_N = C(K_N - N\lambda_0)/\sqrt{N}, \quad \mathcal{G} = \{N(Ah, \Gamma) : h \in H\}.$$

The physical finite statistic is

$$T_{N,J} = Q_J(\tau^{\otimes r}(S_N)), \quad |T_{N,J}| \leq J^r. \quad (8.2)$$

There is no physical jitter, no retained real score, and no omitted overflow event. Every real score belongs to one of the  $J^r$  cells of the open compactified cube, with deterministic boundary tie-breaking.

**Theorem 8.1** (Joint sample and register law). *For the fixed protocol (8.1), there are  $N_0, C_* < \infty$  such that uniformly for  $N \geq N_0$  and  $J \geq 2$ ,*

$$\Delta(\mathcal{E}_{N,J}, \mathcal{G}) \leq C_* \left( N^{-1/2} + \frac{J}{\sqrt{N}} + J^{-2} \right), \quad (8.3)$$



where  $\mathcal{E}_{N,J} = \{\text{Law}_h(T_{N,J}) : h \in H\}$ . If  $AH$  contains an  $r$ -dimensional ball, every experiment with at most  $M$  outputs has distance from  $\mathcal{G}$  at least  $c_* M^{-2/r}$ . Consequently the statistic with  $J = \lfloor M^{1/r} \rfloor$  satisfies

$$c_* M^{-2/r} \leq \Delta(\mathcal{E}_{N,J}, \mathcal{G}) \leq C_{*,K} M^{-2/r}, \quad 2^r \leq M \leq KN^{r/6}, \quad (8.4)$$

for every fixed  $K < \infty$ . In particular  $M \asymp N^{r/6}$  gives distance of order  $N^{-1/3}$ ; achieving distance of order  $N^{-1/5}$  requires and is achieved by a budget of order  $N^{r/10}$ . All constants may depend on the fixed matrices, compact parameter set and covariance conditioning. No uniformity at rank degeneration or unknown marks is asserted.

We prove the uniform estimates in five steps, keeping comparison randomization separate from (8.2).

**Lemma 8.2** (Jittered Poisson local normal comparison). *There is an absolute  $C_P < \infty$  such that for  $\mu \geq 1$ , independent  $U \sim \text{Unif}(-1/2, 1/2)$ , and  $K \sim \text{Pois}(\mu)$ ,*

$$\|\text{Law}(K + U) - N(\mu, \mu)\|_{\text{TV}} \leq C_P \mu^{-1/2}. \quad (8.5)$$

*Proof.* It suffices to consider  $\mu \geq 64$ , increasing the constant for the remaining range. The jittered density is  $p_\mu(k) = e^{-\mu} \mu^k / k!$  on  $[k - 1/2, k + 1/2]$ ,  $k \geq 0$ . On the central cells  $|k - \mu| \leq \mu^{2/3}/4$ , put  $t = k - \mu$ ,  $x = k + u$ ,  $|u| \leq 1/2$ , and  $z = (x - \mu)/\sqrt{\mu}$ . Stirling's formula with remainder  $0 < R_k < 1/(12k)$  gives

$$\log \frac{p_\mu(k)}{\phi_\mu(x)} = -\frac{1}{2} \log(k/\mu) - R_k - \{k \log(k/\mu) - k + \mu\} + \frac{(x - \mu)^2}{2\mu},$$

where  $\phi_\mu$  is the  $N(\mu, \mu)$  density. For  $|t| \leq \mu/2$ , Taylor's theorem applied to  $(1 + v) \log(1 + v) - (1 + v) + 1$  gives remainder at most  $|t|^3/\mu^2$  after its quadratic term. Also  $|\log(1 + t/\mu)| \leq 2|t|/\mu$  and  $|(x - \mu)^2 - t^2| \leq |t| + 1/4$ . Substitution therefore gives the convenient bound

$$\left| \log \frac{p_\mu(k)}{\phi_\mu(x)} \right| \leq 32\mu^{-1/2}(1 + |z|^3), \quad \left| \log \frac{p_\mu(k)}{\phi_\mu(x)} \right| \leq 64. \quad (8.6)$$

The constants are deliberately loose. For example  $|z| \leq \mu^{1/6}/4 + 1/(2\sqrt{\mu})$  on these cells verifies the second bound directly. The inequality  $|e^v - 1| \leq e^{|v|}|v|$  now bounds the integrated central density error by  $32e^{64}\mu^{-1/2}(1 + \mathbb{E}|G|^3)$ ,  $G \sim N(0, 1)$ .

For completeness the omitted tails are exponentially smaller. For  $|s| \leq 1$ , the centered Poisson log moment generating function is  $\mu(e^s - 1 - s) \leq \mu s^2$ . Exponential Markov with  $s = \mu^{-1/3}/16$ , and its negative, bounds the probability outside the central cells, allowing the half-unit jitter margin, by  $2 \exp(-\mu^{1/3}/128)$  after adjusting an absolute factor for  $64 \leq \mu$  in the bounded initial interval. The Gaussian bound follows by its moment generating function. Thus an absolute choice such as  $C_P = 10^8 e^{64}$  dominates the central error, the tails and the finite initial range. This proves a uniform  $\mu^{-1/2}$  bound rather than an unquantified local-limit assertion.  $\square$

**Lemma 8.3** (Variance replacement and projected comparison). *Let  $U_j$  be independent uniform half-cell jitters, used only in this proof, and set  $S_N^o = S_N + CU/\sqrt{N}$ . Uniformly for  $h \in H$ ,*

$$\|\text{Law}_h(S_N^o) - N(Ah, \Gamma)\|_{\text{TV}} \leq C_0 N^{-1/2}. \quad (8.7)$$

One may choose  $C_0$  as a function only of  $m$ , the minimum of  $\lambda_0$ ,  $\max_{h \in H} \|Bh\|$ , and fixed matrix norms.

*Proof.* Put  $b_* = \max_{h \in H} \|Bh\|_\infty$  and  $\lambda_* = \min_j \lambda_{0,j}$ . Choose  $N_0 \geq \max(128/\lambda_*, 4b_*^2/\lambda_*^2)$ . Then each count mean is at least  $N\lambda_*/2$ . Product comparison using Lemma 8.2, followed by centering

and scaling, gives distance at most  $mC_P\sqrt{2/\lambda_*}N^{-1/2}$  from the normal vector with mean  $Bh$  and covariance  $\Lambda + N^{-1/2}\text{diag}(Bh)$ .

For the replacement of this covariance by  $\Lambda$ , the relative variance eigenvalues are  $1 + t_j$ , with  $|t_j| \leq b_*/(\lambda_*\sqrt{N}) \leq 1/2$ . The exact Gaussian relative entropy is  $\frac{1}{2}\sum_j\{t_j - \log(1 + t_j)\} \leq \sum_j t_j^2$ . The elementary entropy-variation inequality  $\|P - Q\|_{\text{TV}} \leq \sqrt{D_{\text{KL}}(P\|Q)/2}$  gives a further bound  $\sqrt{m/2}b_*/(\lambda_*\sqrt{N})$ . Taking  $C_0 = mC_P\sqrt{2/\lambda_*} + \sqrt{m/2}b_*/\lambda_*$  is sufficient. Projection by  $C$  contracts total variation and sends the fixed Gaussian covariance to  $\Gamma$ , proving the assertion.  $\square$

**Lemma 8.4** (Deterministic cell crossing). *Let  $H_0$  be the transformed marginal bound in (7.5), now for the means  $AH$ , and put  $d_N = \|C\|_{\infty,1}/(2\pi\sqrt{N})$ . Then*

$$\sup_{h \in H} \|\text{Law}_h(T_{N,J}) - Q_J \tau^{\otimes r} N(Ah, \Gamma)\|_{\text{TV}} \leq 2C_0 N^{-1/2} + 2rH_0 J d_N. \quad (8.8)$$

Here  $\|C\|_{\infty,1}$  is the maximum absolute row sum.

*Proof.* The coupled scores satisfy  $\|S_N - S_N^o\|_{\infty} \leq \|C\|_{\infty,1}/(2\sqrt{N})$ . Since  $|\tau'| \leq 1/\pi$ , their compactified coordinates differ by at most  $d_N$ . Distinct grid labels require a coordinate of the jittered score to be within  $d_N$  of a grid boundary. On the torus there are  $J$  boundaries per coordinate; under the transformed Gaussian their total probability is at most  $2rH_0 J d_N$ , or its truncation by one. Replacing the jittered law by that Gaussian adds at most  $C_0 N^{-1/2}$ , by (8.7). The coupling inequality then compares theunjittered and jittered labels. One more data-processing comparison of the jittered labels to the Gaussian labels adds the second  $C_0 N^{-1/2}$ . Boundary tie events are already in the strips. This argument never asserts that the unjittered real score has a Gaussian law in total variation.  $\square$

*Proof of Theorem 8.1.* Use the deterministic Gaussian-to-cell kernel for the direction  $\mathcal{G} \rightarrow \mathcal{E}_{N,J}$ . Lemma 8.4 bounds its error by  $b_{N,J} = 2C_0 N^{-1/2} + 2rH_0 J d_N$ . For the reverse direction, use the positive kernel  $H_J$  from Lemma 7.3. Kernel contraction and the triangle inequality give

$$\sup_h \|H_J \text{Law}_h(T_{N,J}) - N(Ah, \Gamma)\|_{\text{TV}} \leq b_{N,J} + rH_2/(12J^2).$$

This proves (8.3), with, for example,  $C_* = 2C_0 + rH_0\|C\|_{\infty,1}/\pi + rH_2/12$  increased to one. Every kernel is parameter independent. Jitter was introduced only to bound a difference of laws, not to modify the physical statistic.

If  $AH$  contains a ball, restrict to an affine parameter section over a smaller mean cube and apply Lemma 7.2. More generally one may simply index the Gaussian family by its mean and choose a Borel representative preimage for each mean in the cube (the compact fibres of the continuous map on  $H$  admit a measurable section); the lower comparison uses only that subfamily. For  $J = \lfloor M^{1/r} \rfloor$  and  $M \geq 2^r$ ,  $J^{-2} \leq 4M^{-2/r}$ . If  $M \leq KN^{r/6}$ , then  $M^{3/r}/\sqrt{N} \leq K^{3/r}$ ; hence  $J/\sqrt{N}$  and  $N^{-1/2}$  are also at most fixed multiples of  $M^{-2/r}$ . This proves the window. Taking  $M$  respectively of order  $N^{r/6}$  and  $N^{r/10}$  gives the two rates. Conversely, an error at most  $CN^{-1/5}$  forces  $M \geq cN^{r/10}$  by the universal cardinality lower bound.  $\square$

**8.1. The A2 matrices and the scope of the consequence.** In A2's known-mark local protocol, the moment derivative  $D$  has  $\Sigma = D\Lambda D^T > 0$ , and its calibrated exposure direction is  $B = \Lambda D^T \Sigma^{-1}$ , so  $DB = I$ . If  $L$  maps to the contact coordinates and  $F$  is an orthonormal frame of its range, the matrices above are

$$C = F^T L D, \quad A = C B = F^T L, \quad \Gamma = C \Lambda C^T > 0.$$

Here  $A$  has full row rank  $r$ . A compact local parameter neighborhood therefore has an image containing a mean ball. Both sides of Theorem 8.1 apply to that actual protocol. This invokes a

structural experiment-cardinality theorem verbatim, not just an analogy between a root singularity and a contact observation.

The result is a matched budget theorem in the displayed joint window. It is not a lower bound saying that  $N^{-1/3}$  is the best sample-size comparison achievable with arbitrarily many labels. It also does not assign a stochastic noise law to a conductor scheme or establish uniformity at a rank-changing physical stratum. The geometric source, the exposure protocol, the finite statistic and the comparison kernels are distinct typed objects throughout.

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